Theory Note 5 of 8

Windows Basics for a SOC Analyst

Network Traffic Analysis, Nmap and Windows Logging | Week 2

1. Overview

Most of the alerts a SOC analyst reviews come from Windows machines, so a working knowledge of how Windows organises itself is essential. The five building blocks below (processes, services, the registry, accounts and UAC) are the places attackers run code, stay persistent and escalate privileges. The three main event logs are where the evidence of all that gets recorded.

2. Core components

Component	What it is	Why an analyst cares
Processes	A running instance of a program, with its own process ID (PID), memory and a parent process. Visible in Task Manager, tasklist or Get-Process.	Check parent-child relationships (for example winword.exe starting powershell.exe), programs running from Temp or AppData, and look-alike names such as svch0st.exe. Event 4688 records process creation.
Services	Programs that run in the background under the Service Control Manager. Start type can be Automatic, Manual or Disabled, and they run as SYSTEM, Local Service or a named user. Managed with services.msc, sc or Get-Service.	Malware often installs a service to survive reboots. Look for unfamiliar names and odd binary paths. Event 7045 (System log) and 4697 (Security log) record installs.
Registry	A hierarchical database of system and user settings. The main hives are HKLM, HKCU, HKCR, HKU and HKCC. Edited with regedit or the reg command.	Persistence often hides in the Run and RunOnce keys and in the Services key. It also holds traces of user activity useful in forensics.
Users and groups	Local accounts and groups such as Administrators, Users and Remote Desktop Users, each identified by a SID. The built-in Administrator SID ends in -500. Managed with lusrmgr.msc or net user.	Account creation (4720), enabling (4722), password resets (4724) and additions to Administrators (4732) are classic signs of privilege abuse.
UAC (User Account Control)	A safeguard that runs even administrators with a standard token and asks for consent or credentials before an action needs elevation.	It shows which actions required elevation. A UAC that has been turned down or bypassed is a red flag. Event 4672 shows when special privileges are assigned at logon.

3. The three main Event Viewer logs

Log	What it records	Examples
Application	Events raised by installed software and applications.	Application crashes (event 1000), antivirus messages, database errors.
Security	Audit events: logons, privilege use, object access, and account and policy changes. Needs an audit policy to be enabled and is readable by administrators only.	4624, 4625, 4672, 4688, 4720, 1102.
System	Events from the operating system and its components, such as drivers, services, start-up and shutdown.	7045 (new service), 7036 (service state change), 6005 and 6006 (event log start and stop), 41 (unexpected reboot).

- Open Event Viewer by running eventvwr.msc, and use Filter Current Log to search by event ID.

- Other logs worth knowing: Microsoft-Windows-PowerShell/Operational (event 4104 records script blocks) and Sysmon/Operational if Sysmon is installed.

> **Key takeaway**
> Processes, services, registry keys and accounts are where attackers leave changes, and the Security and System logs are where those changes are recorded. Learn the normal state of each and the exceptions become obvious.